Ch 7.1

① $A^TA = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ upper

$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ lower

$\sigma_1 = \sqrt{1} = 1$

$\sigma_2 = \sqrt{64} = 8$

$\lambda = 0, 1, 64$ for both

right singular vect: eigvec of A^TA

left " " AA^T

→ $v_1 = (0,0,1)$ $\lambda = 64$

$v_2 = (0,1,0)$, $\lambda = 1$

$u_1 = (0,1,0)$, $\lambda = 64$

$u_2 = (1,0,0)$, $\lambda = 1$

③ $Av = \sigma u$

$\frac{Av}{\sigma} = u \Rightarrow \begin{bmatrix} 5 & 4 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

deepseek
Teslamisola
7.1 #3: SVD

11:23 Tue, 8 Apr • 97%

SVD for Transpose Matrix Singular Vectors

To find the left singular vectors $\mathbf{u}$ and right singular vectors $\mathbf{v}$ for the transpose matrix $A^T = \begin{bmatrix} 5 & 0 \\ 4 & 3 \end{bmatrix}$, we start by noting that the singular values of A^T are the same as those of the original matrix $A = \begin{bmatrix} 5 & 4 \\ 0 & 3 \end{bmatrix}$. The singular vectors switch roles: the left singular vectors of A^T are the right singular vectors of A, and the right singular vectors of A^T are the left singular vectors of A.

First, we compute the SVD of the original matrix A:

1. Compute A^TA and AA^T:
   - $A^TA = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$
   - $AA^T = \begin{bmatrix} 41 & 12 \\ 12 & 9 \end{bmatrix}$
2. Find eigenvalues and eigenvectors of A^TA:
   - Eigenvalues: 45 and 5
   - Corresponding eigenvectors (right singular vectors $\mathbf{v}$ of \(A \):
     - For eigenvalue 45: $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$
     - For eigenvalue 5: $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$
3. Find left singular vectors $\mathbf{u}$ of A using $A\mathbf{v} = \sigma\mathbf{u}$:
   - $\begin{bmatrix} \frac{3}{\sqrt{2}} \end{bmatrix}$

For $\sigma_1 = 3\sqrt{5}$: $\begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}$

- For $\sigma_2 = \sqrt{5}$: $\begin{bmatrix} \frac{1}{\sqrt{10}} \\ -\frac{3}{\sqrt{10}} \end{bmatrix}$

For the transpose matrix A^T, the left singular vectors $\mathbf{u}$ are the right singular vectors of A, and the right singular vectors $\mathbf{v}$ are the left singular vectors of A:

- Left singular vectors $\mathbf{u}$ of A^T:
  - $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$
  - $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$
- Right singular vectors $\mathbf{v}$ of A^T:
  - $\begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}$
  - $\begin{bmatrix} \frac{1}{\sqrt{10}} \\ -\frac{3}{\sqrt{10}} \end{bmatrix}$

Final Answer

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4 deepseek sharad33 · · · 7.1 #4

11:41 Tue, 8 Apr • 96%

Symmetric Matrix Reveals SVD Eigenvalues

> And we need to show that S has eigenvalues σ_k and -σ_k, where σ_k are the singular values of A. The given equations are Av_k = σ_k u_k and A^T u_k = σ_k v_k. These look familiar because they are part of the singular value decomposition (SVD) of A. So, u_k

and v_k are the left and right singular vectors of A, respectively.

The problem also mentions that the eigenvectors of S are [u_k; v_k] and [-u_k; v_k]. Let me verify this. If I apply S to the first eigenvector [u_k; v_k], what happens?

Multiplying S with [u_k; v_k]:

First block row: 0u_k + Av_k = Av_k. But from the given equation, Av_k = σ_k u_k.

Second block row: A^Tu_k + 0v_k = A^Tu_k. From the given equation, A^Tu_k = σ_k v_k.

So putting it together, S * [u_k; v_k] = [σ_k u_k; σ_k v_k] = σ_k [u_k; v_k]. Therefore, [u_k; v_k] is indeed an eigenvector with eigenvalue σ_k.

Now, for the second eigenvector [-u_k; v_k], let's do the same multiplication:

First block row: 0*(-u_k) + Av_k = Av_k = σ_k u_k (from before). But here, the first component of the eigenvector is -u_k, so the result of the first block is σ_k u_k. However, the eigenvector's first component is -u_k, so we need to see if this equals -σ_k*(-u_k) or something else. Wait, maybe I need to compute it step by step.

Wait, let's compute S * [-u_k; v_k]:

First block: 0*(-u_k) + Av_k = Av_k = σ_k u_k.

Second block: A^T*(-u_k) + 0*v_k = -A^T u_k = -σ_k v_k (since A^T u_k = σ_k v_k).

So the result is [σ_k u_k; -σ_k v_k]. Now, compare this to -σ_k times the eigenvector [-u_k; v_k]:

-σ_k * [-u_k; v_k] = [σ_k u_k; -σ_k v_k], which matches exactly. Therefore, S * [-u_k; v_k] = -σ_k * [-u_k; v_k], so [-u_k; v_k] is an eigenvector with eigenvalue -σ_k.

So that shows that S has eigenvalues ±σ_k. Also, since S is a symmetric matrix (because S^T = [0 A^T; A 0]^T = [0 A; A^T 0] = S), its eigenvalues are real, which makes sense because singular values σ_k are non-negative real numbers, so their negatives are also real.

Additionally, the trace of S is mentioned to be zero. The trace of a matrix is the sum of its diagonal elements. In S, the diagonal blocks are zero matrices, so the trace is indeed 0. Since the trace is also the sum of the eigenvalues, this implies that the sum of all eigenvalues (σ_k and -σ_k) is zero. That makes sense because for each σ_k, there is a corresponding -σ_k, so they cancel out when summed.

This setup connects the singular values and vectors of A to the eigenvalues and eigenvectors of the larger matrix S. So instead of working directly with A, which might not be symmetric, we can study the symmetric matrix S to reveal the SVD of A. This is useful because symmetric matrices have orthogonal eigenvectors and real eigenvalues, which are properties that can be leveraged in computations and proofs.

I need to make sure there are no mistakes here. Let me check again:

1. Applied S to both eigenvectors and confirmed they result in σ_k and -σ_k times the eigenvectors. That seems correct.

2. Trace of S is sum of diagonals, which are zero, so trace is zero. Sum of eigenvalues (including both σ_k and -σ_k) must also be zero. That checks out.
3. S is symmetric because swapping blocks and transposing gives the original matrix. Correct.

Therefore, the explanation holds.

The matrix $S = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix}$ is symmetric and connects to the Singular Value Decomposition (SVD) of A. Here's the explanation:

1. Eigenvectors and Eigenvalues:
   - For the eigenvector $\begin{bmatrix} u_k \\ v_k \end{bmatrix}$:

$$S \begin{bmatrix} u_k \\ v_k \end{bmatrix} = \begin{bmatrix} Av_k \\ A^T u_k \end{bmatrix} = \begin{bmatrix} \sigma_k u_k \\ \sigma_k v_k \end{bmatrix} = \sigma_k \begin{bmatrix} u_k \\ v_k \end{bmatrix}.$$

     This shows σ_k is an eigenvalue.
   - For the eigenvector $\begin{bmatrix} -u_k \\ v_k \end{bmatrix}$:

$$S \begin{bmatrix} -u_k \\ v_k \end{bmatrix} = \begin{bmatrix} Av_k \\ -A^T u_k \end{bmatrix} = \begin{bmatrix} \sigma_k u_k \\ -\sigma_k v_k \end{bmatrix} = -\sigma_k \begin{bmatrix} -u_k \\ v_k \end{bmatrix}.$$

     This shows $-\sigma_k$ is an eigenvalue.
2. Trace of S:
   - The trace is the sum of diagonal elements. Since the diagonal blocks of S are zero, $\text{Trace}\,(S) = 0$.
   - The eigenvalues σ_k and $-\sigma_k$ sum to zero, consistent with the trace.
3. Significance:
   - S is symmetric, ensuring real eigenvalues and orthogonal eigenvectors.
   - The eigenvectors of S directly correspond to the left (u_k \)) and right (\(v_k \)) singular vectors of \(A , and the eigenvalues are the singular values σ_k and their negatives.

Conclusion: The symmetric matrix S encapsulates the SVD of A, with eigenvalues $\pm\sigma_k$ and eigenvectors containing the singular vectors of A. The trace condition confirms the eigenvalue pairing (σ_k and \(-\sigma_k\) sums to zero.

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(5) $A^TA = \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix}$

$\lambda^2 - 25\lambda - 0 = 0$

$\lambda(\lambda - 25) = 0 \quad \therefore \lambda = 25, 0$

$\begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ $\quad 2x + y = 0$

$v_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$2x = -y$

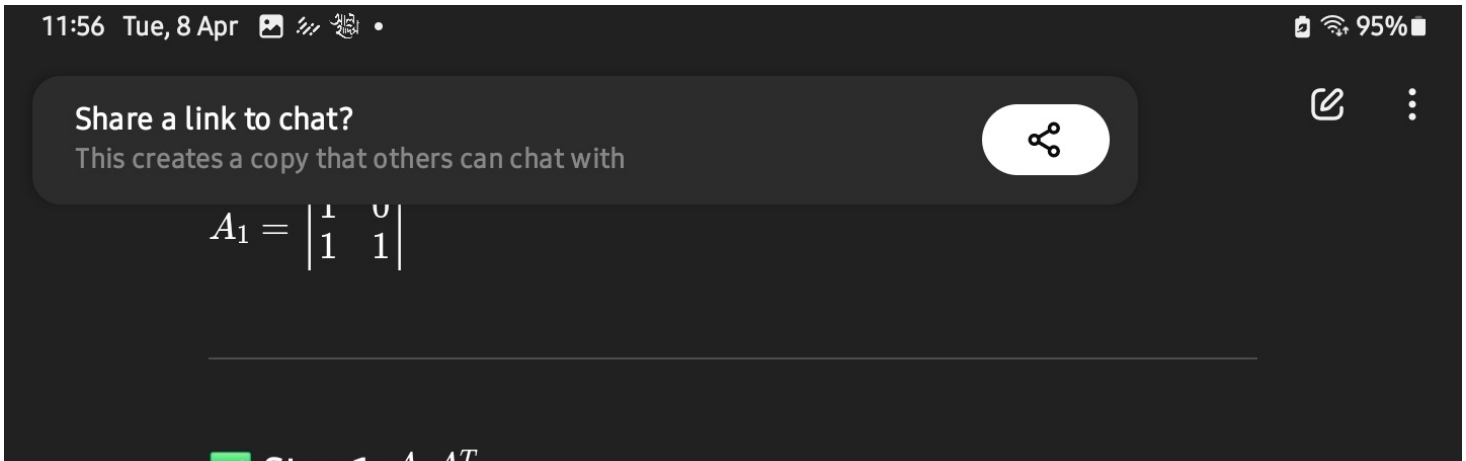
$\begin{bmatrix} -5 & 10 \\ 10 & -20 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \quad -5x = -10y$

$x = 2y$

$v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

v_1 & v_2 are ortho.

(6)



✅ Step 1: $A_1 A_1^T$

$$A_1 A_1^T = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$$

✅ Step 2: Eigenvalues

Characteristic polynomial:

$$\lambda^2 - 3\lambda + 1 = 0 \Rightarrow \lambda_{1,2} = \frac{3 \pm \sqrt{5}}{2}$$

Approximations:

- $\lambda_1 \approx 2.618$
- $\lambda_2 \approx 0.382$

So singular values are:

$$\sigma_1 = \sqrt{2.618} \approx 1.618, \quad \sigma_2 = \sqrt{0.382} \approx 0.618$$

✅ Step 3: Eigenvectors → Columns of U

Solve $(A_1 A_1^T - \lambda I)u = 0$

For $\lambda_1 = 2.618$, matrix becomes:

$$\begin{vmatrix} 1 - 2.618 & 1 \\ 1 & 2 - 2.618 \end{vmatrix} = \begin{vmatrix} -1.618 & 1 \\ 1 & -0.618 \end{vmatrix}$$

Solving gives eigenvector:

$$u_1 \propto \begin{vmatrix} 1 \\ 1.618 \end{vmatrix}, \quad \text{normalize: } \frac{1}{\sqrt{1^2 + (1.618)^2}} = \frac{1}{\sqrt{3.618}} \approx 0.525$$

So:

$$u_1 \approx 0.525 \begin{vmatrix} 1 \\ 1.618 \end{vmatrix} \approx \begin{vmatrix} 0.525 \\ 0.850 \end{vmatrix}$$

For $\lambda_2 = 0.382$, similar steps give:

$$u_2 \approx \begin{vmatrix} -0.851 \\ 0.526 \end{vmatrix}$$

Thus:

$$U \approx \begin{vmatrix} 0.525 & -0.851 \\ 0.850 & 0.526 \end{vmatrix}$$

✅ Step 4: Σ

$$\Sigma = \begin{vmatrix} 1.618 & 0 \\ 0 & 0.618 \end{vmatrix}$$

✅ Step 5: Compute V from $v_i = \frac{A^T u_i}{\sigma_i}$

Use:

$$v_1 = \frac{A^T u_1}{\sigma_1}, \quad v_2 = \frac{A^T u_2}{\sigma_2}$$

Compute:

- $A^T u_1 = \begin{bmatrix} 1 & 1\,0 & 1 \end{bmatrix} \begin{bmatrix} 0.525\ 0.850 \end{bmatrix} = \begin{bmatrix} 1.375\ 0.850 \end{bmatrix}$
- Divide by $\sigma_1 \approx 1.618$:
  $v_1 \approx \begin{bmatrix} 0.850\ 0.526 \end{bmatrix}$

Similarly:

- $A^T u_2 = \begin{bmatrix} 1 & 1\,0 & 1 \end{bmatrix} \begin{bmatrix} -0.851\ 0.526 \end{bmatrix} = \begin{bmatrix} -0.325\ 0.526 \end{bmatrix}$
- Divide by $\sigma_2 \approx 0.618$:
  $v_2 \approx \begin{bmatrix} -0.526\ 0.851 \end{bmatrix}$

Thus:

$$V \approx \begin{vmatrix} 0.850 & -0.526 \\ 0.526 & 0.851 \end{vmatrix} \quad \Rightarrow \quad V^T \approx \begin{vmatrix} 0.850 & 0.526 \\ -0.526 & 0.851 \end{vmatrix}$$

✅ Final Answer (rounded)

SVD of $A_1 = \begin{bmatrix} 1 & 0\,1 & 1 \end{bmatrix}$ is:

$$U \approx \begin{bmatrix} 0.525 & -0.851 \\ 0.850 & 0.526 \end{bmatrix}, \quad \Sigma \approx \begin{bmatrix} 1.618 & 0 \\ 0 & 0.618 \end{bmatrix}, \quad V^T \approx \begin{bmatrix} 0.850 & 0.526 \\ -0.526 & 0.851 \end{bmatrix}$$

If you want the full steps for A_2 as well, I got you! Just say the word, akhi.

You need GPT-4o to continue this chat because there's an attachment. Your limit resets in 4 hours.

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8

12:07 Tue, 8 Apr • 94%

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Ah, I see where you're coming from. You're correct that the singular values of $A+I$ are related to the square roots of the eigenvalues of $(A+I)^T(A+I)$, but let's clarify this further in detail.

Singular Values and Eigenvalues:

1. Singular Values of $A+I$: The singular values of a matrix M are the square roots of the eigenvalues of the matrix M^TM. So, for $A+I$, the singular values are the square roots of the eigenvalues of $(A+I)^T(A+I)$.

$$\sigma_i(A+I) = \sqrt{\lambda_i\left((A+I)^T(A+I)\right)}$$

1. Why Not A^TA+I? The singular values of $A+I$ are not the square roots of the eigenvalues of A^TA+I because these are different matrices and reflect different transformations. Here's why:
   - A^TA is the Gram matrix of A, and its eigenvalues are related to the squared singular values of A.
   - $(A+I)^T(A+I)$ represents the Gram matrix of $A+I$, and it has a structure that depends on both A and I. The addition of I (which is a shift by an identity matrix) alters the eigenvalues in a way that doesn't correspond simply to adding I to A^TA.

   Therefore, A^TA+I does not have the same spectral properties as $(A+I)^T(A+I)$, and so the singular values of $A+I$ cannot be determined from A^TA+I.

Key Differences:

- $(A+I)^T(A+I) = A^TA + A + A^T + I$, which is a different matrix than $A^TA + I$.
- The matrix $A^TA + I$ doesn't capture the full structure of $(A+I)^T(A+I)$, so the eigenvalues (and thus singular values) of these matrices will differ.

Conclusion:

The singular values of $A+I$ are related to the eigenvalues of $(A+I)^T(A+I)$, not $A^TA + I$. The latter doesn't directly correspond to the singular values of $A+I$ because the structure of the two matrices differs. This is why ; not directly compute the singular values of $A+I$ by modi...g the SVD of A in a simple way.

Ask anything

9) $A = Q$ → orthogonal Mat.

Singular values of of Q are + sqrt of eigvals (λ) of Q^TQ and all those " are 1. cz

$Q^TQ = I$ when Q orthogonal.

so, every singular val of $Q = 1$.

10

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Given:

- A symmetric matrix $S = A^TA$, where A has eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, all positive.
- The eigenvectors $v_1, v_2, \ldots, v_n$ are orthonormal (i.e., $v_i^Tv_j = \delta_{ij}$,

where δ_{ij} is the Kronecker delta).

- Any vector x can be written as a linear combination of the eigenvectors:

$$x = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

$$R(x) = \frac{\lambda_1 c_1^2 + \lambda_2 c_2^2 + \cdots + \lambda_n c_n^2}{c_1^2 + c_2^2 + \cdots + c_n^2}$$

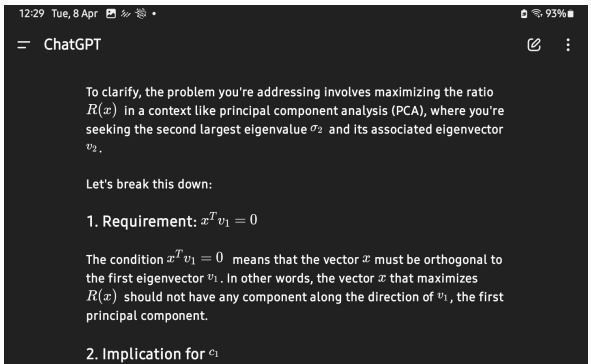
Maximizing and Minimizing $R(x)$

- To maximize $R(x)$: We observe that $R(x)$ is a weighted average of the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. The maximum value of $R(x)$ occurs when all the weight is on the largest eigenvalue λ_1 . This corresponds to choosing the vector $x = v_1$, the eigenvector corresponding to λ_1 .
  - For this case, $c_1 = 1$ and $c_2 = c_3 = \cdots = c_n = 0$.
  - The maximum value of $R(x)$ is λ_1 .
- To minimize $R(x)$: Similarly, the minimum value of $R(x)$ occurs when all the weight is on the smallest eigenvalue λ_n . This corresponds to choosing the vector $x = v_n$, the eigenvector corresponding to λ_n .
  - For this case, $c_1 = c_2 = \cdots = c_{n-1} = 0$ and $c_n = 1$.
  - The minimum value of $R(x)$ is λ_n .

Summary:

- The vector that maximizes $R(x)$ is $x = v_1$, and the maximizing coefficients are $c_1 = 1, c_2 = c_3 = \cdots = c_n = 0$. The maximum value of $R(x)$ is λ_1 .
- The vector that minimizes $R(x)$ is $x = v_n$, and the minimizing coefficients are $c_1 = c_2 = \cdots$ $_{-1} = 0, c_n = 1$. The minimum value of $R(x)$ is λ_n .

Ask anything



In terms of the linear combination $x = c_1v_1 + c_2v_2 + \cdots + c_nv_n$, the condition $x^Tv_1 = 0$ implies that the coefficient c_1, which corresponds to the first eigenvector v_1, must be zero. This is because the inner product of x with v_1 is:

$$x^Tv_1 = c_1v_1^Tv_1 + c_2v_2^Tv_1 + \cdots + c_nv_n^Tv_1$$

Since the eigenvectors are orthogonal, $v_i^Tv_j = 0$ for $i \neq j$, and $v_1^Tv_1 = 1$. Thus, we have:

$$x^Tv_1 = c_1 = 0$$

3. Maximizing the Ratio for σ_2 at v_2

The objective is to maximize the ratio:

$$\frac{\lambda_2c_2^2+\lambda_3c_3^2+\cdots+\lambda_nc_n^2}{c_2^2+c_3^2+\cdots+c_n^2}$$

Here, $\lambda_2, \lambda_3, \ldots, \lambda_n$ are the eigenvalues associated with the eigenvectors $v_2, v_3, \ldots, v_n$.

The maximum of this ratio occurs when all the weight is placed on the second largest eigenvalue λ_2, corresponding to the eigenvector v_2, i.e., when $c_2 = 1$ and $c_3 = c_4 = \cdots = c_n = 0$. This results in:

$$\frac{\lambda_2}{1} = \lambda_2$$

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principal component.
2. Implication for c_1
In terms of the linear combination $x = c_1v_1 + c_2v_2 + \cdots + c_nv_n$, the condition $x^Tv_1 = 0$ implies that the coefficient c_1, which corresponds to the first eigenvector v_1, must be zero. This is because the inner product of x with v_1 is:
$x^Tv_1 = c_1v_1^Tv_1 + c_2v_2^Tv_1 + \cdots + c_nv_n^Tv_1$
Since the eigenvectors are orthogonal, $v_i^Tv_j = 0$ for $i \neq j$, and $v_1^Tv_1 = 1$. Thus, we have:
$x^Tv_1 = c_1 = 0$
3. Maximizing the Ratio for σ_2 at v_2
The objective is to maximize the ratio:
$\frac{\lambda_2c_2^2+\lambda_3c_3^2+\cdots+\lambda_nc_n^2}{c_2^2+c_3^2+\cdots+c_n^2}$
Here, $\lambda_2, \lambda_3, \ldots, \lambda_n$ are the eigenvalues associated with the eigenvectors $v_2, v_3, \ldots, v_n$.
The maximum of this ratio occurs when all the weight is placed on the second largest eigenvalue λ_2, corresponding to the eigenvector v_2, i.e., when $c_2 = 1$ and $c_3 = c_4 = \cdots = c_n = 0$. This results in:
$\frac{\lambda_2}{1} = \lambda_2$

So, $\sigma_2 = \lambda_2$, and the second eigenvector v_2 is the direction that maximizes $R(x)$ under the constraint that x is orthogonal to v_1.

Summary:

- The condition $x^Tv_1 = 0$ forces $c_1 = 0$ in the linear combination $x = c_1v_1 + c_2v_2 + \cdots + c_nv_n$.
- The maximization of the ratio $R(x)$ at $x = v_2$ implies that the coefficient $c_2 = 1$ and all other coefficients $c_3, c_4, \ldots, c_n$ are 0.
- This leads to $\sigma_2 = \lambda_2$, where λ_2 is the second largest eigenvalue.

Ask anything

(12) $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$ $A^TA = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$

$\lambda^2 - 10\lambda + 16 \therefore \lambda = 8, 2$ $\sigma = \sqrt{8}, \sqrt{2}$

$\begin{bmatrix} 5-8 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

$$\begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix}$$

$$v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

↓ normalize

$$\begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

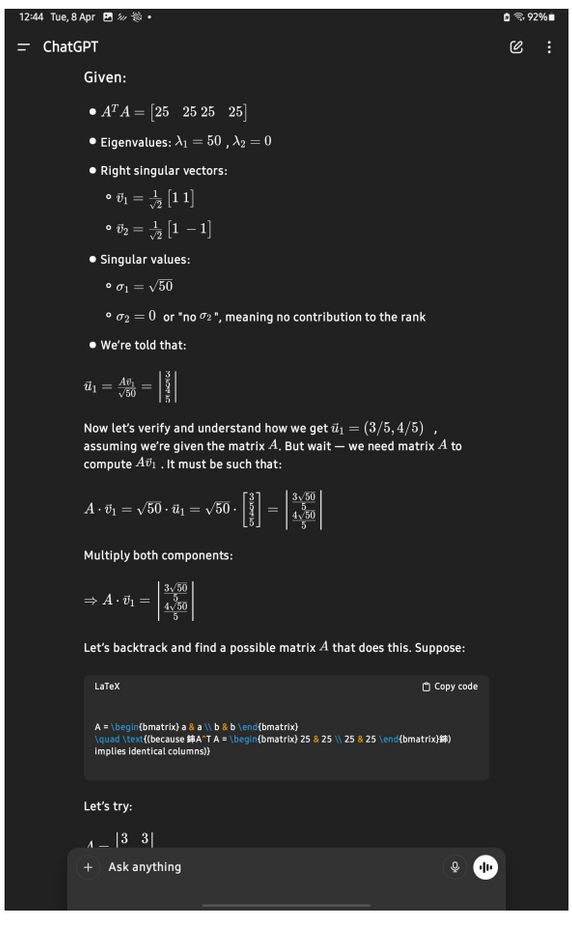
$$v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u = \frac{Av}{\sigma} = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$$

$$6-8 = 2\sqrt{2}$$

$$\frac{\frac{1}{\sqrt{2}}\begin{bmatrix} 4 \\ 0 \end{bmatrix}}{2\sqrt{2}}$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



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$$A\vec{v}_1 = A \cdot \frac{1}{\sqrt{2}} \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \frac{1}{\sqrt{2}} \cdot \begin{vmatrix} 2+2 \\ -1+1 \end{vmatrix} = \frac{1}{\sqrt{2}} \begin{vmatrix} 4 \\ 0 \end{vmatrix}$$

Now divide by $\sigma_1 = 2\sqrt{2}$:

$$\vec{u}_1 = \frac{1}{2\sqrt{2}} \begin{vmatrix} 4 \\ 0 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

---

2. Compute $\vec{u}_2$

Use $\vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \end{bmatrix}$,
and $\sigma_2 = \sqrt{2}$

$$A\vec{v}_2 = A \cdot \frac{1}{\sqrt{2}} \begin{vmatrix} 1 \\ -1 \end{vmatrix} = \frac{1}{\sqrt{2}} \cdot \begin{vmatrix} 2-2 \\ -1-1 \end{vmatrix} = \frac{1}{\sqrt{2}} \begin{vmatrix} 0 \\ -2 \end{vmatrix}$$

Now divide by $\sigma_2 = \sqrt{2}$:

$$\vec{u}_2 = \frac{1}{\sqrt{2}} \begin{vmatrix} 0 \\ -2 \end{vmatrix} \cdot \frac{1}{\sqrt{2}} = \begin{vmatrix} 0 \\ -1 \end{vmatrix}$$

Sometimes the sign doesn't matter for singular vectors (orthonormal basis is still valid), so you may see:

$$\vec{u}_2 = \begin{vmatrix} 0 \\ 1 \end{vmatrix}$$

Final Result:

- Right singular vectors $\vec{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}$, $\vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \end{bmatrix}$
- Singular values: $\sigma_1 = 2\sqrt{2}$, $\sigma_2 = \sqrt{2}$
- Left singular vectors:

$$\vec{u}_1 = \begin{vmatrix} 1 \\ 0 \end{vmatrix}, \quad \vec{u}_2 = \begin{vmatrix} 0 \\ 1 \end{vmatrix}$$

Let me know if you want the full SVD breakdown next!

Ask anything

12:50 Tue, 8 Apr 91%

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Matrix A is:

$$A = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

And you're told:

- A^T A = \begin{bmatrix}1 & 1 & 0 \\1 & 2 & 1 \\0 & 1 & 1\end{bmatrix}
- Eigenvalues of $A^T A$: $\lambda_1 = 3, \lambda_2 = 1, \lambda_3 = 0$
- So the singular values of A are:
  - $\sigma_1 = \sqrt{3}$
  - $\sigma_2 = 1$
  - $\sigma_3 = 0$ (→ no third singular value since A is 2×3)

Eigenvectors of $A^T A$ (i.e., right singular vectors v_i):

- $v_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$
- $v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$
- $v_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}$ no need just show.

These form the columns of matrix V in the SVD.

To find the left singular vectors u_i, you use:

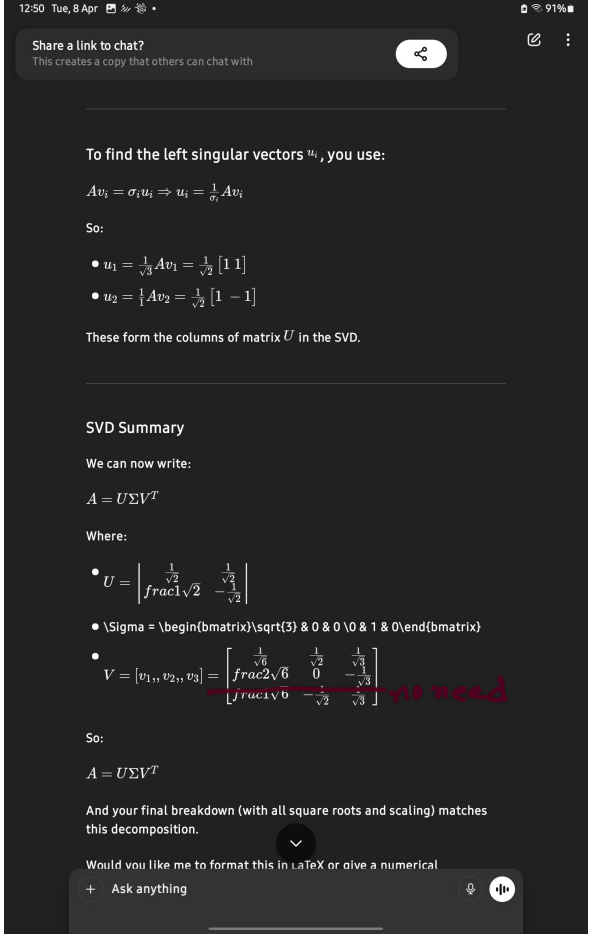
$$Av_i = \sigma_i u_i \Rightarrow u_i = \frac{1}{\sigma_i} Av_i$$

So:

- $u_1 = \frac{1}{\sqrt{3}} Av_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}$
- $u_2 = \frac{1}{1} Av_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \end{bmatrix}$

These form the columns of matri he SVD.

Ask anything



(14) $A^TA = \sigma^2 v$

$= \lambda v$

$A(A^TA) = A\lambda v$

$\Rightarrow (AA^T)A = \lambda(Av)$ [Associativity]

So if v is an Eigvec of A^TA

then Av " " " AA^T

eigval same λ

12:59 Tue, 8 Apr 90%

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🌟 The Key Idea:

If

$(A^TA)v = \lambda v$

$(AA^T)(Av) = \lambda(Av)$

This means:

> The image of an eigenvector of A^TA, when multiplied by A, becomes an eigenvector of AA^T with the same eigenvalue.

📌 So What Does This Imply?

1. Connection Between Two Symmetric Matrices:

- A^TA and AA^T are both symmetric and positive semi-definite.
- They share the same nonzero eigenvalues.
- Their eigenvectors are related via the transformation A and A^T.

---

2. Foundation of Singular Value Decomposition (SVD):

The SVD of a matrix A is:

$A = U\Sigma V^T$

Where:

- V: orthonormal eigenvectors of A^TA
- U: orthonormal eigenvectors of AA^T
- Σ: diagonal matrix of singular values (square roots of the eigenvalues λ)

This whole setup depends on the fact that if

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- U: orthonormal eigenvectors of AA^T
- Σ: diagonal matrix of singular values (square roots of the eigenvalues λ)

This whole setup depends on the fact that if

$A^TAv = \lambda v \Rightarrow Av$ is an eigenvector of AA^T

So you get the vectors $u_i = \frac{Av_i}{|Av_i|}$ as left singular vectors.

---

3. Geometric Meaning:

This shows how a transformation A affects space:

- It stretches or squashes the input vector v (right singular vector),
- and sends it into the direction of Av (left singular vector),
- scaling it by the singular value $\sigma = \sqrt{\lambda}$.

✅ In Summary:

This result implies:

- The right singular vectors of A are eigenvectors of A^TA.
- The left singular vectors of A are eigenvectors of AA^T.
- The nonzero eigenvalues of both are the squares of the singular values of A.
- The matrix A links these two spaces together via the transformation:

$Av_i = \sigma_i u_i$

It's the mathematical bridge between domain and codomain in linear transformations!

Ask anything

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Suppose A = UTVT is 2 by 2 with g; > g2 > 0. Change A by as small a matrix

as possible to produce a singular matrix Ag. Hint: U and V' do not change.soln: 18 The smallest change in A is to set its smallest singular value σ2 to zero

You're given a 2×2 matrix $A = U\Sigma V^T$, where $\Sigma = \begin{bmatrix}\sigma_1 & 0\\ 0 & \sigma_2\end{bmatrix}$, and $\sigma_1 > \sigma_2 > 0$. This is the Singular Value Decomposition (SVD) of A, where:

- U and V are orthogonal matrices (2×2),
- Σ contains the singular values σ_1 and σ_2,
- You want to make as small a change as possible to A to make it singular, i.e., make $\det(A) = 0$.

🧠 Key Insight:

A matrix is singular if it loses rank—i.e., has determinant zero, or equivalently, at least one of its singular values is zero.

So the smallest possible change is:

✅ Set the smallest singular value σ_2 to zero.

Let's define the modified matrix A_g as:

$$A_g = U\Sigma_g V^T, \quad \text{where} \quad \Sigma_g = \begin{vmatrix}\sigma_1 & 0\\ 0 & 0\end{vmatrix}$$

So the change is:

$$\Delta A = A_g - A = U(\Sigma_g - \Sigma)V^T = U \begin{vmatrix}0 & 0\\ 0 & -\sigma_2\end{vmatrix} V^T$$

This is the smallest change in A (in terms of 2-norm) that makes it singular, and it only removes the contribution of the smallest singular value σ_2 .

Let me know if you want a numerical example or geometric intuition for this!

Ask anything

17

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Inverse of Matrix and Singular Values Analysis

> Final Answer
>
> The inverse of is . Therefore, .

Given a square invertible matrix A with its singular value decomposition (SVD) as $A = UZV^T$, where U and V are orthogonal matrices and Z is the diagonal matrix of singular values, we need to find the inverse of A.

1. Inverse of A:
   - The inverse of A can be derived using its SVD. Since $A = UZV^T$, the inverse A^{-1} is given by:
   $$A^{-1} = (UZV^T)^{-1} = VZ^{-1}U^T$$
   - This is because the inverse of an orthogonal matrix is its transpose, and the inverse of a diagonal matrix Z is another diagonal matrix with the reciprocals of the singular values.
2. Largest Singular Value of A^{-1}:
   - The singular values of A are the diagonal entries of Z. The singular values of A^{-1} are the reciprocals of these values. The largest singular value of A^{-1} is therefore $1/\sigma_{\min}(A)$, where $\sigma_{\min}(A)$ is the smallest singular value of A.
3. Largest Eigenvalue of A^{-1}:
   - The eigenvalues of A^{-1} are the reciprocals of the eigenvalues of A. The largest eigenvalue in magnitude of A^{-1} is $1/|\lambda_{\min}(A)|$, where $|\lambda_{\min}(A)|$ is the smallest eigenvalue of A in magnitude.
4. Relationship Between Singular Values and Eigenvalues:
   - The smallest singular value of A is less than or equal to the smallest

• The smallest singular value of A is less than or equal to the smallest eigenvalue of A in magnitude, i.e., $\sigma_{\min}(A) \leq |\lambda_{\min}(A)|$. This is derived from the fact that the spectral radius (largest eigenvalue magnitude) of A^{-1} is less than or equal to the spectral norm (largest singular value) of A^{-1}.

5. Condition Number:

• The condition number of A, which is the ratio of the largest singular value to the smallest singular value (\sigma_{\text{max}}/\sigma_{\text{min}} \)), is equal to \(\sigma_{\text{max}}(A^{-1}) \cdot \sigma_{\text{max}}(A) . This ratio is always greater than or equal to 1 and equals 1 if A is a multiple of an orthogonal matrix.

Final Answer

The inverse of A is $VZ^{-1}U^T$. Therefore, $A^{-1} = \boxed{V}\boxed{Z^{-1}}\boxed{U^T}$.

New chat

Message DeepSeek

DeepThink (R1) Search

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.

Let's break down the points from your question:

(a) Trace of A^TA :

The matrix A^TA is a square matrix, and its main diagonal elements are the squared lengths of the rows of the matrix A.

- If A is an $m \times n$ matrix, the element a_{ij} represents the entry in the i-th row and j-th column of A.
- The squared length of the i-th row of A is given by the sum of the squares of its elements:

$\|\text{row}_i\|^2 = \sum_{j=1}^{n} a_{ij}^2$

$(A^TA)_{ii} = \|\text{row}_i\|^2 = \sum_{j=1}^{n} a_{ij}^2$

$\text{Tr}(A^TA) = \sum_{i=1}^{m} \|\text{row}_i\|^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2$

(b) If A has rank 1:

When A has rank 1, it means that all the rows (or columns) of A are linearly dependent and can be written as scalar multiples of each other. Therefore, A can be represented as the outer product of two vectors u and v, i.e., $A = uv^T$.

- Since A has rank 1, the matrix A^TA will also have rank 1.
- The singular values of A are the square roots of the eigenvalues of A^TA .
- The matrix A^TA will have only one non-zero eigenvalue (since its rank is 1), and this eigenvalue is equal to the trace of A^TA because the trace of a matrix is the sum of its eigenvalues.
- Thus, the only non-zero singular value σ_1 of A is the square root of the trace of A^TA :

$$\sigma_1 = \sqrt{\mathrm{Tr}(A^TA)}$$

This explains why $\sigma_1 = \left(\mathrm{Tr}(A^TA)\right)^{1/2}$ when A has rank 1.

Ask anything

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$$A = \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix}$$

We compute A^TA :

$$A^T = \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix}$$

Then:

$$A^TA = \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = \begin{vmatrix} 10 & 20 \\ 20 & 40 \end{vmatrix}$$

Now, we find the eigenvalues of A^TA by solving the characteristic equation:

$$\det(A^TA - \lambda I) = 0$$

$$\det \begin{vmatrix} 10-\lambda & 20 \\ 20 & 40-\lambda \end{vmatrix} = 0$$

The determinant of a 2x2 matrix is:

$$\det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Thus, we get:

$$(10-\lambda)(40-\lambda) - 20 \times 20 = 0$$

Expanding this:

$$(10-\lambda)(40-\lambda) - 400 = 0$$

$$400 - 10\lambda - 40\lambda + \lambda^2 - 400 = 0$$

$$\lambda^2 - 50\lambda = 0$$

$$\lambda(\lambda - 50) = 0$$

The eigenvalues are:

$$\lambda_1 = 0, \quad \lambda_2 = 50$$

The singular values are the square roots of the eigenvalues:

Ask anything

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The right singular vectors ... and ... are the eigenvectors of A^TA .

For $\lambda_1 = 0$, the eigenvector corresponding to this eigenvalue is the null space of A^TA . Similarly, for $\lambda_2 = 50$, we solve for the eigenvector corresponding to this eigenvalue.

Eigenvector for $\lambda_2 = 50$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad A^TA = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}$$

$$\lambda^2 - 50\lambda = 0 \therefore \lambda(\lambda - 50) \therefore \lambda = 0, 50$$

$$AA^T = \begin{bmatrix} 5 & 15 \\ 15 & 45 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$5x = -15y$$

$$x = -3y$$

$$\frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

U2 honey!

We solve $(A^TA - 50I)v = 0$:

$$\begin{vmatrix} 10-50 & 20 \\ 20 & 40-50 \end{vmatrix} \begin{vmatrix} v_1 \\ v_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

This gives:

$$\begin{vmatrix} -40 & 20 \\ 20 & -10 \end{vmatrix} \begin{vmatrix} v_1 \\ v_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

Solving this system, we get:

$$v_1 = \begin{vmatrix} 1 \\ 2 \end{vmatrix}$$

Eigenvector for $\lambda_1 = 0$:

The eigenvector corresponding to $\lambda_1 = 0$ is the null space of A^TA, which leads to:

$$v_2 = \begin{vmatrix} -2 \\ 1 \end{vmatrix}$$

Step 4: Compute the Left Singular Vectors (u_1 and u_2)

The left singular vectors u_1 and u_2 are the eigenvectors of AA^T. To compute them, we use the relation:

$$u_1 = \frac{Av_1}{\sigma_1}, \quad u_2 = \frac{Av_2}{\sigma_2}$$

Since $\sigma_2 = 0$, we only compute u_1 :

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{50}} A \begin{vmatrix} 1 \\ 2 \end{vmatrix}$$

This gives:

Ask anything

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Step 4: Compute the Left Singular Vectors (u_1 and u_2)

The left singular vectors u_1 and u_2 are the eigenvectors of AA^T. To compute them, we use the relation:

$$u_1 = \frac{Av_1}{\sigma_1}, \quad u_2 = \frac{Av_2}{\sigma_2}$$

Since $\sigma_2 = 0$, we only compute u_1 :

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{50}} A \begin{vmatrix} 1 \\ 2 \end{vmatrix}$$

This gives:

$$u_1 = \frac{1}{\sqrt{50}} \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} \begin{vmatrix} 1 \\ 2 \end{vmatrix} = \frac{1}{\sqrt{50}} \begin{vmatrix} 5 \\ 15 \end{vmatrix}$$

Simplifying:

$$u_1 = \frac{1}{\sqrt{50}} \begin{bmatrix} 5 \\ 15 \end{bmatrix} = \begin{vmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{vmatrix}$$

Since $\sigma_2 = 0$, u_2 is the null vector:

$$u_2 = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

Final Results:

- Right singular vectors:
  - $v_1 = \begin{bmatrix} 1 & 2 \end{bmatrix}$
  - $v_2 = \begin{bmatrix} -2 & 1 \end{bmatrix}$
- Left singular vectors:
  - $u_1 = \begin{vmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{vmatrix}$
  - $u_2 = \begin{bmatrix} 0 & 0 \end{bmatrix}$

These are the singular vectors corresponding to the matrix A in the decomposition $A = U\Sigma V^T$.

A=UΣV

Ask anything

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- Σ is a diagonal matrix containing the singular values.

The decomposition is shown as:

$$A = U\Sigma V^T$$

Where:

- $U = \begin{bmatrix} u_1 & u_2 \end{bmatrix}$
- $\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 \end{bmatrix}$
- $V = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$

From the provided information:

- The row space has basis v_1.
- The null space has basis v_2.
- The column space has basis u_1.
- The left null space has basis u_2.

The SVD breakdown is:

- $\Sigma = \sqrt{10} \begin{bmatrix} \sqrt{50} & 0 & 0 & 0 \end{bmatrix}$
- $U = \begin{bmatrix} 1 & 3 & 3 & -1 \end{bmatrix}$
- $V = \begin{bmatrix} 1 & 2 & 2 & -1 \end{bmatrix}$

Interpretation of the Spaces:

- Row space has basis v_1, which is $\sqrt{\frac{1}{5}} \begin{bmatrix} 1 & 2 \end{bmatrix}$.
- Null space has basis v_2, which is $\sqrt{\frac{1}{5}} \begin{bmatrix} -2 & 1 \end{bmatrix}$.
- Column space has basis u_1, which is $\sqrt{\frac{1}{10}} \begin{bmatrix} 1 & 3 \end{bmatrix}$.
- Left null space has basis u_2, which is $\sqrt{\frac{1}{10}} \begin{bmatrix} -3 & 1 \end{bmatrix}$.

This decomposition provides the singular values and the corresponding left and right singular vectors, as ...s the relationships between the row space, null space, column sp... left null space of the matrix A.

Ask anything

No, the specific relationships you've mentioned between the row space, null space, column space, and left null space are not true for all matrices. They are valid for matrices that have full row or column rank, but they depend on the rank of the matrix and its specific properties.

Let's break it down:

1. Row Space (Basis v_1)

The row space of a matrix is the subspace spanned by the rows of the matrix. In SVD, v_1 (the first right singular vector) is related to the row space. Specifically:

- The right singular vector v_1 corresponds to the first non-zero singular value σ_1, and it lies in the row space of A.
- If the matrix has full rank, v_1 will be non-zero and will span the row space.

For your example:

- The basis of the row space is indeed $v_1 = \begin{bmatrix} 1 & 2 \end{bmatrix}$, which has been normalized.

2. Null Space (Basis v_2)

The null space of a matrix consists of all vectors x such that $Ax = 0$. In SVD, the right singular vector v_2 (corresponding to σ_2) is related to the null space of A. Specifically:

- If $\sigma_2 = 0$ (i.e., the matrix has a zero singular value), v_2 is in the null space of A.
- v_2 spans the null space of A if A is rank-deficient.

For your example:

- The basis of the null space is $v_2 = \begin{bmatrix} -2 & 1 \end{bmatrix}$, normalized to $\frac{1}{\sqrt{5}}$, because $\sigma_2 = 0$.

3. Column Space (Basis u_1)

The column space is the span of t umns of the matrix. In the SVD,

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3. Column Space (Basis u_1)

The column space is the span of the columns of the matrix. In the SVD, u_1 (the first left singular vector) corresponds to the first non-zero singular value σ_1 and lies in the column space of A. Specifically:

- u_1 represents the first direction in the column space.
- If the matrix has full column rank, u_1 will span the column space.

For your example:

- The basis of the column space is $u_1 = \begin{bmatrix} 1 & 3 \end{bmatrix}$, normalized to $\frac{1}{\sqrt{10}}$.

4. Left Null Space (Basis u_2)

The left null space is the null space of A^T, i.e., the set of vectors y such that $A^T y = 0$. In SVD, u_2 (the second left singular vector) corresponds to σ_2, and if $\sigma_2 = 0$, u_2 represents the direction orthogonal to the column space of A. Specifically:

- If $\sigma_2 = 0$, the left null space is non-trivial, and u_2 spans it.

For your example:

- The basis of the left null space is $u_2 = \begin{bmatrix} -3 & 1 \end{bmatrix}$, normalized to $\frac{1}{\sqrt{10}}$.

Generalizing:

These relationships are true for any matrix but depend heavily on the rank of the matrix and the non-zero singular values:

- If the matrix has full rank (no zero singular values), then u_2 will typically not be the zero vector, and the matrix will have linearly independent left and right singular vectors.
- If the matrix is rank-deficient, u_2 and v_2 might become zero vectors, as in the case where $\sigma_2 = 0$, indicating the presence of a non-trivial null space.

Thus, while the general structure is correct, the specific details, like u_2, depend on the properties of the matrix A and its rank.

Ask anything